A09: Independent Adversarial Audit
Transcritical Bifurcation at the Origin

Directed Cyclic Networks research project

3 October 2026 Audited version: P09 v1.0

CRITICAL SPECTRUM / REDUCTION VERDICT:	**VERIFIED**
GENERIC TRANSCRITICAL VERDICT:	**VERIFIED**
STABILITY / PHYSICAL-BRANCH VERDICT:	**VERIFIED**
GENERIC CRITICAL-DYNAMICS VERDICT:	**VERIFIED**
DEGENERATE CRITICAL-STABILITY VERDICT:	**VERIFIED**
CODIMENSION / SCOPE VERDICT:	**VERIFIED**
OVERALL VERDICT:	**VERIFIED**

Required mathematical or wording corrections: none. The exact submitted source proves the generic full-system local transcritical bifurcation and the separate cubic-degenerate result within its stated local domains. In particular, the full origin at $b = \mu = 0$ is locally asymptotically stable in $\mathcal{U}$, and is not locally exponentially stable. The highest-priority Lyapunov remainder and quadratic-form arguments are valid as written; their details are reconstructed below, independently of the developer's checker.

This is an audit verdict, not a lifecycle edit. P09 and every existing project file remain unchanged. No P10 proof is begun. The original P09 development chat owns administrative closure and the complete audited project archive.

1 Exact current-file authentication and audit contract

The authoritative input is `cyclic_network_project_P09_draft.zip`. The manifest's CURRENT P09 entry is at lines 693–732, with the exact full-file hash at line 730. It supersedes historical TODO/not-started records. The audited file is precisely `cyclic_network_project/proofs/P09_transcritical_bifurcation.tex`, P09 v1.0, dated 3 October 2026: 24,155 bytes and 539 lines.

Full P09 SHA-256:
`b2366edf1ff71199b7f1d02a58cf828148d9a697dfafc9e07e70a523e4f26680`

Mathematical body SHA-256, line 17 through EOF:
`a0830944d46d26de35faf09c5f0e4d9f42d6f10620ca22373f6bcf7031705393`

Original complete draft ZIP SHA-256:
`61df24eb644771cbb84027447c4a21ba1c1d0e1d44dc8c35cf34a0a0c755c8de`

The two existing draft ZIP copies are byte-identical. The archive passes its CRC check, contains exactly 143 files under the sole root `cyclic_network_project/`, and every member matches the located `output/`

`p09/cyclic_network_project/` working copy. Thus no earlier placeholder or reconstructed source is being audited. The current-file check was completed before accepting any mathematical conclusion.

All ten requested control files were located and authenticated: manifest, claim registry, frozen theorem package, final novelty verdict, novelty decisions, dependency map, compendium source, shared macros, canonical model, and notation/assumptions. Current P01–P05, P07, P08 and their A01–A05, A07, A08 sources were also located. The additional open-decisions file was consulted for B03 and NB07. Full hashes, sizes and line counts for these 26 inputs are in `A09_support/input_inventory.json`; all 143 archive-member hashes are in `A09_support/archive_member_hashes.json`.

Each current prerequisite body hash matches its header's declared current body hash. Comparison against the complete P07-audited baseline confirms that all prerequisite proofs and prior audits are unchanged; the only five changed members in the P09 draft are P09 itself, manifest, registry, compendium source and compendium PDF. A01/A02 E01 and A05/A08 E01 are already closed as recorded; A03/A04/A07 require no correction. A02's restored audit source is present: its historical source-absence header is explicitly superseded by the current manifest. A04/A05 historical numerical-provenance qualifications remain nonblocking and are not silently declared resolved here.

The frozen documents retain historical status prose. Their current scope is consistent with the CURRENT P09 entry: supporting local branch organization; no two-parameter unfolding; no new priority claim. Exact-model priority remains unclosed. This audit performed no new literature survey.

2 Dependency authentication

The frozen mathematical edges remain

$$\mathrm{P02}, \mathrm{P03}, \mathrm{P04} \longrightarrow \mathrm{P09}, \qquad \mathrm{P01 \text{ inherited.}}$$

The following are the actual imported facts, with their original hypotheses.

Input	Use and scope checked
P01	Smooth local field and continuation on $\mathcal{U} = (-1, \infty)^n$; nonnegative invariance when restricting full-domain stability to $\mathbb{R}^n_{\geq 0}$. The canonical positive tuple satisfies P01's weaker hypotheses.
P02	Exact invariant synchrony and scalar restriction (invariance proposition and scalar equation); algebraic repeated-root locus (loci proposition). No global nonsynchronous classification is imported.
P03	Original Jacobian and critical-origin structure; direct differentiation in this audit verifies the same formula, including additive offset collisions.
P04	Homogeneous circulant coefficients and Fourier spectrum at every synchronous state $r > -1$, even when negative or nonstationary; uniform derivative identity; nonlinear implications of a Hurwitz Jacobian or a positive-real-part eigenvalue.
Standard theory	Finite-dimensional real IFT; fixed-parameter and extended center manifolds; stability reduction with stable complement; strict Lyapunov/continuation arguments; regular level-set theorem. These are ordinary mathematical tools, not extra project-module edges.

The only external proof-label references in P09 are:

`prop:P01-local`, `prop:P02-invariance`, `eq:P02-scalar`, `eq:P03-origin-jacobian`, `eq:P04-synchronous-derivative`, and `prop:P02-loci`.

Every label exists in the authenticated source. Inspection of uncited arguments also finds no reliance on P05, P07, P08 or P10–P18. References to these modules only describe scheduling or exclusions. In particular, transverse stability uses a finite collection of continuous eigenvalue formulas, not the P05 selection theorem. **Dependency verdict: PASS.**

3 Layer A: critical spectrum and real transverse reduction

All statements fix $a, k, c, h > 0$, $n \geq 5$, $1 \leq q < n$. Set

$$A = k + h > 0, \quad b = a - A, \quad g_c = a + c, \quad \mu = g_c - g, \qquad \frac{d\mu}{dg} = -1.$$

Restrict $|\mu| < g_c/2$ so that $g > 0$. The original field is

$$F_i = ax_i - kx_i^2 + \frac{cx_{i-1}}{1 + x_i} - \frac{(g_c - \mu)x_i}{1 + x_{i+q}} - hx_i x_{i-q}. \tag{1}$$

Its denominators do not vanish on $\mathcal{U}$; it is jointly smooth there.

3.1 Independent critical differentiation and gap

At zero, the input numerator derivative contributes c at predecessor column $i - 1$, its denominator derivative vanishes, and the rational loss contributes $-g$ only at column i. Both derivatives of the h product and the quadratic self-loss derivative vanish. Therefore

$$D_x F(0, \mu) = (a - g)I + cP = \mu I + c(P - I), \qquad J_c = c(P - I).$$

All q terms have vanished before any Fourier argument. A disconnected step-q graph cannot create extra kernel vectors because the surviving predecessor shift P is a single n-cycle.

For $v_j^{(m)} = n^{-1/2} e^{\mathrm{i}\theta_m j}$, $Pv^{(m)} = e^{-\mathrm{i}\theta_m} v^{(m)}$; hence

$$\lambda_m^c = c(e^{-\mathrm{i}\theta_m} - 1), \qquad \theta_m = 2\pi m/n.$$

The n vectors form a unitary basis, so this lists algebraic multiplicities. Only $m = 0$ gives zero, and the characteristic polynomial $(\lambda + c)^n - c^n$ has derivative $nc^{n-1} \neq 0$ there. The equation $Pz = z$ forces every coordinate equal, giving real kernel $\mathbb{R}\mathbf{1}$ and geometric multiplicity one.

For $m \neq 0$, $j = \min(m, n - m)$ satisfies $1 \leq j \leq \lfloor n/2 \rfloor$ and $2\pi j/n \in [2\pi/n, \pi]$. Monotonicity of cosine yields

$$\Re\lambda_m^c = -c(1 - \cos\theta_m) \leq -\delta_n, \qquad \delta_n = c(1 - \cos(2\pi/n)) > 0.$$

For even n, the self-conjugate label $m = n/2$ has eigenvalue $-2c$; for odd n, the upper representative angle is strictly below π and the same inequality holds. Equality at the closest nonzero labels is allowed. Neither the formula nor its proof contains q or $\gcd(n, q)$. P09 lines 54–85: **PASS**.

3.2 A real IFT with explicit local quantifiers

Let $w^T = n^{-1}\mathbf{1}^T$, $E = \ker w^T$ and $\Pi = I - \mathbf{1}w^T$. Then $w^T\mathbf{1} = 1$, $\Pi^2 = \Pi$, $\Pi^T = \Pi$ and

$$x \longleftrightarrow (r, y) = (w^T x, \Pi x), \quad (r, y) \longmapsto r\mathbf{1} + y$$

are inverse real linear maps between $\mathbb{R}^n$ and $\mathbb{R} \times E$. In particular $\|x\|_2^2 = nr^2 + \|y\|_2^2$. Since $w^T P = w^T$, J_c preserves E and $w^T J_c = 0$.

Define $T(r, y, \mu) = \Pi F(r\mathbf{1} + y, \mu)$. It maps to the real space E and $D_y T(0, 0, 0) = J_c|_E$. To justify real invertibility without any hidden complex step, if a real $y \in E$ has $J_c y = 0$, the kernel calculation forces $y = t\mathbf{1}$, and $w^T y = 0$ forces $t = 0$. Injectivity on the finite-dimensional real space E gives surjectivity and a real inverse. Equivalently, its complexification has no zero eigenvalue; a singular real restriction would produce a complex kernel vector. Thus P09's complexification argument is sufficient and correct.

The real IFT gives open neighborhoods I_r, I_μ of zero and an open $Y \subset E$ containing zero such that, within $I_r \times Y \times I_\mu$,

$$T(r, y, \mu) = 0 \quad \Longleftrightarrow \quad y = \psi(r, \mu).$$

These sets can be chosen with $r\mathbf{1} + y \in \mathcal{U}$ throughout. Substitution into the original field gives

$$F(r\mathbf{1}, \mu) = f(r, \mu)\mathbf{1}, \qquad f = ar - Ar^2 + \frac{(c-g)r}{1+r} = \frac{r}{1+r}(\mu + br - Ar^2).$$

Consequently $T(r, 0, \mu) = 0$ for *every* $(r, \mu) \in I_r \times I_\mu$. The fixed neighborhood Y contains zero for every such pair; uniqueness forces $\psi \equiv 0$. No state-dependent choice of Y is involved. Taking a smaller open state ball whose (r, y) image lies in $I_r \times Y$ and a smaller parameter interval produces the claimed common product in (x, μ) coordinates.

Every equilibrium in that product is therefore synchronous, and conversely it is an equilibrium exactly when

$$r(\mu + br - Ar^2) = 0.$$

Multiplication by $1 + r > 0$ neither removes nor creates a root; $r = -1$ is excluded by the domain. This proof uses no division by b, so covers $b = 0$. It rules out all nearby nonsynchronous equilibria, not merely smooth nonsynchronous branches. P09 lines 87–130: **PASS**.

CRITICAL SPECTRUM / REDUCTION VERDICT: VERIFIED.

4 Layer B: generic transcritical geometry and dynamics

4.1 Full-vector derivatives, normalization and orientation

The scalar restriction gives $f(0, \mu) = 0$, $f_r(0, \mu) = \mu$ and $f_\mu = r/(1+r)$. Thus $f_r(0, 0) = 0$, $f_{r\mu}(0, 0) = 1$ and, since

$$f(r, 0) = br^2 - (A + b)r^3 + O(r^4),$$

$f_{rr}(0, 0) = 2b$. The quadratic Taylor coefficient is $\frac{1}{2} f_{rr} = b$, not $2b$.

For $v = \mathbf{1}$ and $w^T = n^{-1}\mathbf{1}^T$, $w^T v = 1$, $J_c v = 0$ and $w^T J_c = 0$. Directly in (1),

$$(F_\mu)_i = \frac{x_i}{1 + x_{i+q}}, \quad D_x F_\mu(0) = I, \quad F_\mu(0) = 0.$$

Replacing μ by g reverses the sign, so $w^T F_{x\mu} v = 1$ and $w^T F_{xg} v = -1$.

To check every second directional contribution, substitute $x = tv$ at $g = g_c$ and differentiate each original summand twice at $t = 0$:

summand	at	$-kt^2$	$ct/(1+t)$	$-g_c t/(1+t)$	$-ht^2$
second derivative	0	$-2k$	$-2c$	$2g_c$	$-2h$

Their sum in each of the n components is $2(g_c - c - k - h) = 2b$. Averaging with w cancels the factor n exactly: $w^T F_{xx}[v, v] = 2b$. These are full-vector coefficients, not coefficients inferred from an unverified

scalar reduction. For $b \neq 0$, mixed and quadratic nondegeneracy both hold; at $b = 0$ the second one fails. P09 lines 134–169: **PASS**.

4.2 Unique local nonzero branch and both slopes

When $b \neq 0$, $Q(r, \mu) = Ar^2 - br - \mu$ has $Q_r(0, 0) = -b \neq 0$. A second real IFT gives the unique root $r_*(\mu)$ near zero. $Q(0, \mu) = -\mu$ implies $r_*(\mu) = 0$ only at $\mu = 0$. The other algebraic root tends to $b/A \neq 0$; for example, choose a state radius less than $|b|/(3A)$ and shrink the parameter interval so the other root stays outside. This is a fixed-b statement. No uniform neighborhood as $b \to 0$ is claimed or available.

Implicit differentiation gives

$$(2Ar_* - b)r'_* = 1, \qquad 2A(r'_*)^2 + (2Ar_* - b)r''_* = 0.$$

At zero these yield

$$r'_*(0) = -\frac{1}{b}, \qquad r''_*(0) = \frac{2A}{b^3}, \qquad r_*(\mu) = -\frac{\mu}{b} + \frac{A\mu^2}{b^3} + O(\mu^3).$$

Because $d\mu/dg = -1$, $dr_*/dg(g_c) = 1/b$. In the (r, μ) plane the branch tangents $(0, 1)$ and $(-1/b, 1)$ are independent. P09 explicitly restricts its transverse-crossing claim to that plane; it makes no ambient $(n + 1)$-dimensional transversality claim. P09 lines 174–225: **PASS**.

4.3 Exact center restriction and positive time change

At fixed $\mu = 0$, the invariant synchronous line is tangent to the entire one-dimensional center space, hence is a smooth local center manifold. For the extended system with $\dot{\mu} = 0$, $F_\mu(0, 0) = 0$ implies the linearization $\mathrm{diag}(J_c, 0)$. The synchronous line times the parameter interval is invariant and tangent to its full two-dimensional center space. The remaining $n - 1$ eigenvalues are strictly stable.

For $b \neq 0$, define $u = Ar^2 - br = r(Ar - b)$. Its derivative $2Ar - b$ is nonzero near zero, so $(r, \mu) \mapsto (u, \mu)$ is a local smooth coordinate change on this center manifold. Since $\mu + br - Ar^2 = \mu - u$,

$$\dot{u} = (2Ar - b)\frac{r}{1 + r}(\mu - u) = \alpha(r)u(\mu - u), \quad \alpha(r) = \frac{2Ar - b}{(Ar - b)(1 + r)}.$$

The identity extends smoothly to $r = 0$ because $Ar - b \neq 0$ there and $\alpha(0) = 1$. On $|r| < 1/2$, $2A|r| < |b|/2$, both $Ar - b$ and $2Ar - b$ retain the sign of $-b$, while $1 + r > 0$. Thus $\alpha > 0$ and $d\tau/dt = \alpha(r)$ gives exactly $du/d\tau = u(\mu - u)$. This is an orbital equivalence preserving forward time on the exact center restriction. It does not assert a decoupled equation for arbitrary transverse perturbations. P09 lines 227–248: **PASS**.

With the equilibrium completeness and full stability proof below, the persistent branch, simple center eigenvalue, nonzero coefficients, distinct branch tangents, exact center dynamics and stable transverse complement justify the full-system nondegenerate transcritical classification. This conclusion does not follow from scalar geometry alone, and P09 does not use that invalid shortcut.

GENERIC TRANSCRITICAL VERDICT: VERIFIED.

5 Layer C: full stability exchange and physical branch

5.1 Uniform sign and transverse persistence

Differentiating the scalar field before imposing equilibrium gives

$$f_r = \frac{\mu + br - Ar^2}{(1+r)^2} + \frac{r}{1+r}(b - 2Ar).$$

At a nonzero equilibrium $\mu = -r_*(b - Ar_*)$, so

$$\lambda_0^{\mathrm{nz}} = \frac{r_*}{1+r_*}(b - 2Ar_*) = -\mu \frac{b - 2Ar_*}{(1+r_*)(b - Ar_*)}.$$

For $|r_*| < 1/2$ and $2A|r_*| < |b|/2$, both factors involving b have the sign of b, and the displayed fraction is positive. Thus

$$\operatorname{sgn} \lambda_0^{\mathrm{nz}} = -\operatorname{sgn} \mu$$

exactly, on either side, for either sign of b. Its expansion $-\mu + O(\mu^2)$ corroborates this sign and is not used as the main proof. P09 lines 250–270: **PASS**.

Independent differentiation of the original field at $r\mathbf{1}$ gives

$$\begin{aligned} J(r,\mu) &= dI + CP + GP^{-q} - HP^q, \\ d &= a - 2kr - \frac{cr}{(1+r)^2} - \frac{g_c - \mu}{1+r} - hr, \\ C &= \frac{c}{1+r}, \qquad G = \frac{(g_c - \mu)r}{(1+r)^2}, \qquad H = hr. \end{aligned}$$

Coincident columns must be added, as in the defining field. Hence $\lambda_m = d + Ce^{-\mathrm{i}\theta_m} + Ge^{\mathrm{i}q\theta_m} - He^{-\mathrm{i}q\theta_m}$. These coefficients remain real and smooth at negative $r > -1$; positivity of G and H is not being assumed. Each nonzero label has limit λ_m^c at $(0,0)$. Intersecting finitely many continuity neighborhoods gives a common neighborhood where

$$\Re\lambda_m(r,\mu) < -\delta_n/2 \quad (m \neq 0).$$

This also holds when $b = 0$; the proof never divided by b. P09 lines 272–292: **PASS**.

5.2 Nonlinear conclusions and number of unstable directions

At the origin, $\lambda_0 = \mu$ and $\Re\lambda_m = \mu - c(1 - \cos\theta_m)$; for $|\mu| < \delta_n/2$ all nonuniform modes remain strictly negative. Intersect this interval with the reduction, branch and gap neighborhoods. For $\mu \neq 0$ all equilibria under discussion are hyperbolic. P04's checked nonlinear linearization implications apply to the smooth field on its open domain:

Parameter side	Origin	Local nonzero branch
$\mu < 0$, $g > g_c$	Locally exponentially stable	Unstable
$\mu > 0$, $g < g_c$	Unstable	Locally exponentially stable

The unstable uniform eigenvalue is real and simple, and all transverse eigenvalues have negative real parts. Thus the unstable spectral space, and the local unstable manifold, have real dimension one at each unstable *off-critical* equilibrium in this table. P09's "here" at line 300 has precisely that local proof context. It does not ascribe a positive eigenvalue to the nonhyperbolic unstable origin discussed later at $\mu = 0$, nor assert that every perturbation of an unstable point escapes. No revision of this wording is required. P09 lines 294–303: **PASS**.

5.3 Positive side and actual nonnegative escape

The relation $\mu = -r_*(b - Ar_*)$, with the second factor retaining the sign of b, proves $r_* > 0$ exactly when $\operatorname{sgn}\mu = -\operatorname{sgn} b$ for nonzero local μ. Therefore:

- $b < 0$ ($a < k + h$): the positive branch lies on $\mu > 0$, or $g < g_c$, and is locally exponentially stable; the origin is unstable.
- $b > 0$ ($a > k + h$): the positive local branch lies on $\mu < 0$, or $g > g_c$, and is unstable; the origin is locally exponentially stable.

This last branch is the small root, not any other positive root separated from zero for fixed $b > 0$.

At each fixed $\mu > 0$, $f(r, \mu)/r \to \mu$ as $r \downarrow 0$. Choose an $\varepsilon > 0$ inside the local domain with $f(r, \mu) \geq (\mu/2)r$ on $(0, \varepsilon]$. A solution with arbitrarily small $r_0 > 0$ is increasing and must reach ε by time at most $2\mu^{-1}\log(\varepsilon/r_0)$: otherwise the integrated inequality would contradict $r < \varepsilon$. Continuation holds until exit from the compact scalar interval. The exact lift $r(t)\mathbf{1}$ is nonnegative throughout. This proves origin instability relative to $\mathbb{R}^n_{\geq 0}$ by an admissible trajectory, not merely a positive eigenvector. The fixed escape radius may depend on μ, as the local instability definition allows.

For $\mu < 0$, full-domain exponential stability and P01's nonnegative invariance restrict to the orthant. A positive unstable branch point is interior to the orthant; its positive scalar derivative also supplies small synchronous departures remaining positive until a fixed local exit. P09 lines 305–330: **PASS**.

STABILITY / PHYSICAL-BRANCH VERDICT: VERIFIED.

6 Layer D: generic critical-point dynamics and state-space scope

At $\mu = 0$, $b \neq 0$, the invariant-line equation is exactly

$$\dot{r} = \frac{r^2(b - Ar)}{1 + r}.$$

Choose $0 < \varepsilon < 1/2$ and $A\varepsilon < |b|/2$. For $0 < |r| \leq \varepsilon$, $b - Ar$ has the sign of b and magnitude at least $|b|/2$, whereas $0 < 1 + r \leq 3/2$. Thus $|\dot{r}| \geq |b|r^2/3$. If $b > 0$, take $r_0 > 0$; if $b < 0$, take $r_0 < 0$. In both cases $s = |r|$ increases and

$$\dot{s} \geq \frac{|b|}{3}s^2, \qquad \frac{1}{s(t)} \leq \frac{1}{s(0)} - \frac{|b|}{3}t$$

until $s = \varepsilon$. The exit time is at most $3|b|^{-1}(s(0)^{-1} - \varepsilon^{-1})$. Smooth continuation within $[-\varepsilon, \varepsilon] \subset (-1, \infty)$ rules out any earlier failure of existence. These arbitrarily small data exit one fixed full-state neighborhood because $\|r\mathbf{1}\|_2 = \sqrt{n}|r|$.

This proves nonlinear instability in the open domain $\mathcal{U}$ in both generic cases. For $b > 0$ the escaping states are nonnegative. For $b < 0$ they are negative, and the proof does not imply relative-orthant instability. P09 explicitly makes no full relative-orthant critical stability claim in that latter case; no subsequent sentence enlarges it. Statements of orthant instability elsewhere have $\mu > 0$, or $b > 0, \mu = 0$, as their hypotheses. P09 lines 332–353: **PASS**.

GENERIC CRITICAL-DYNAMICS VERDICT: VERIFIED.

7 Layer E: degenerate geometry, spectra and center stability

Set $b = 0$, hence $a = A = k + h$. The transverse IFT already proved local completeness without this genericity assumption. The remaining equation is

$$f(r, \mu) = \frac{r}{1+r}(\mu - Ar^2).$$

For $\mu < 0$ there is no nonzero real root; for $\mu = 0$ only the origin; for $\mu > 0$ the two nonzero roots are exactly $r_\pm = \pm\sqrt{\mu/A}$. For $0 < \mu < A$ both satisfy $r > -1$, and a further parameter shrinkage puts both in every required reduction and spectral-gap neighborhood. Only the positive one is nonnegative. There are no hidden nonsynchronous equilibria.

At the critical point,

$$f(r, 0) = -\frac{Ar^3}{1+r}, \qquad f_r(0,0) = f_{rr}(0,0) = 0, \qquad f_{rrr}(0,0) = -6A \neq 0.$$

"Triple zero" is the multiplicity of the scalar field's zero, not a triple eigenvalue. The full Jacobian is still J_c with one simple zero. At either nonzero root,

$$\lambda_0 = f_r(r_\pm, \mu) = -\frac{2Ar_\pm^2}{1+r_\pm} < 0.$$

The negative root is strictly greater than -1, so its denominator is positive; the numerator is negative on *both* branches. Combining with the same transverse gap proves full nonlinear local exponential stability of both roots in $\mathcal{U}$. The origin is exponentially stable for $\mu < 0$ and unstable for $\mu > 0$, with the earlier synchronous positive escape proving relative-orthant instability. Both nonzero roots occur on $g < g_c$, because $\mu = g_c - g$. P09 lines 358–403: **PASS**.

7.1 The center-manifold theorem is applicable as written

The reduction principle being used is the fixed-equilibrium, finite-dimensional center-stability principle: for a sufficiently smooth local system whose linearization splits into a center spectrum on the imaginary axis and a strictly stable spectrum, with no unstable spectrum, local asymptotic stability of the equilibrium in the restriction to a local center manifold implies local asymptotic stability of the full equilibrium. A smooth invariant manifold through the equilibrium tangent to the complete center subspace is a valid local center manifold here.

At the fixed tuple $b = \mu = 0$, the original field is smooth, the invariant synchronous line is tangent to $\mathbb{R}\mathbf{1}$, the remaining eigenvalues have real part at most $-\delta_n$, and no unstable spectrum exists. On that line,

$$\frac{d}{dt}\frac{r^2}{2} = -\frac{Ar^4}{1+r} < 0 \quad (0 < |r| < 1/2).$$

The scalar origin is locally asymptotically stable from both sides. All hypotheses of the stated reduction principle hold. P09 explicitly fixes $\mu = 0$ and excludes the frozen parameter direction from this stability application. The extended two-dimensional center manifold used for the generic normal form is not being misused here. The application in P09 lines 405–415 is rigorous as written; the direct proof below is independent confirmation, not a repair of a defective theorem application.

8 Highest-priority audit: direct full-system Lyapunov argument

Everything in this section is at the fixed degenerate critical tuple, with $g = g_c$, $a = A$. Use the real coordinates $x = r\mathbf{1} + y$, $y \in E$.

8.1 Why the mean remainder is quadratic in the transverse state

Let $J(r) = D_x F(r\mathbf{1}, 0)$. Independently from the original field, $J(r) = d(r)I + C(r)P + G(r)P^{-q} - H(r)P^q$, even at a nonstationary synchronous state. Its column sums, as well as row sums, equal $d + C + G - H = f_r(r, 0)$. Therefore

$$w^T J(r) y = f_r(r, 0) w^T y = 0 \qquad \text{for every } r \text{ and } y \in E.$$

This is an identity at each r, not just at $r = 0$. In particular it eliminates any putative mixed $r\,y$ term in the mean equation.

Choose a compact convex coordinate neighborhood whose image lies in $\mathcal{U}$, so the entire segment $r\mathbf{1} + ty$, $0 \le t \le 1$, lies there. Taylor's formula in the transverse variable gives

$$\begin{aligned} F(r\mathbf{1} + y, 0) = F(r\mathbf{1}, 0) + J(r)y \\ + \int_0^1 (1 - t) D_x^2 F(r\mathbf{1} + ty, 0)[y, y]\, dt. \end{aligned}$$

After averaging, only the first and last terms remain. The second derivatives are bounded uniformly on this compact set; hence

$$\dot{r} = -\frac{Ar^3}{1 + r} + R(r, y), \qquad |R(r, y)| \le K \|y\|_2^2.$$

The constant K is chosen on an initial neighborhood and kept fixed while the later radius shrinks. There is no circular choice of K and ρ. P09 lines 419–426: **PASS**.

For an additional independent algebraic check, put $s = 1 + r$ and $\langle z_i \rangle = n^{-1} \sum_i z_i$. Expanding the original rational terms exactly gives

$$\begin{aligned} R = & -k\langle y_i^2 \rangle - h\langle y_i y_{i-q} \rangle \\ & + c \left\langle \frac{r y_i^2}{s^2(s + y_i)} - \frac{y_{i-1} y_i}{s(s + y_i)} \right\rangle \\ & - g_c \left\langle \frac{r y_{i+q}^2}{s^2(s + y_{i+q})} - \frac{y_i y_{i+q}}{s(s + y_{i+q})} \right\rangle. \end{aligned} \tag{2}$$

This follows from the exact identity

$$\frac{r + p}{s + z} = \frac{r}{s} + \frac{p}{s} - \frac{rz}{s^2} + \frac{rz^2}{s^2(s + z)} - \frac{pz}{s(s + z)}.$$

All linear terms average to zero because each cyclic shift of y has zero mean. For $|r|, \|y\|_2 \le 1/4$, $s \ge 3/4$ and $s + y_i \ge 1/2$. The two rational coefficients are bounded by $8/9$ and $8/3$ respectively. Cauchy–Schwarz for each shifted product yields the explicit sufficient constant

$$|R| \le K_0 \|y\|_2^2, \qquad K_0 = \frac{1}{n} \left[A + \frac{32}{9}(c + g_c) \right].$$

This extra identity is audit evidence only; P09's bounded-Taylor proof already suffices. It works when offsets collide or n, q are noncoprime.

8.2 Exact transverse factorization

Since $\Pi F(r\mathbf{1}, 0) = 0$, the fundamental theorem of calculus gives

$$\dot{y} = \Pi \int_0^1 D_x F(r\mathbf{1} + ty, 0) y\, dt = J_c y + B(r, y) y,$$

where, as an operator from E to E,

$$B(r,y) = \Pi \int_0^1 [D_x F(r\mathbf{1} + ty, 0) - J_c]\, dt \Big|_E .$$

Here $\Pi J_c y = J_c y$. This is exact, including $y = 0$, and not an approximate factorization requiring division by $\|y\|$. Smoothness implies $B(0,0) = 0$ and continuity in operator norm, so $\|B(r,y)\|_2 \le \delta_n/2$ on a sufficiently small neighborhood. P09 lines 427–434: **PASS**.

8.3 The quadratic-form estimate does not rely on nonnormal spectra

For a real vector y, the skew-symmetric part contributes zero:

$$y^T J_c y = y^T S y, \qquad S = \frac{J_c + J_c^T}{2} = c\left(\frac{P + P^{-1}}{2} - I\right).$$

The real symmetric matrix S is unitarily diagonalized by the same Fourier basis, with eigenvalues $c(\cos\theta_m - 1)$. On E its uniform Fourier coefficient is zero; by Parseval,

$$y^T S y = \sum_{m=1}^{n-1} c(\cos\theta_m - 1)|\widehat{y}_m|^2 \le -\delta_n \sum_{m=1}^{n-1} |\widehat{y}_m|^2 = -\delta_n \|y\|_2^2.$$

This includes the real singleton when n is even. The bound is sharp on the real sine/cosine space of labels $1, n-1$. P09's short phrase "Fourier diagonalization gives" refers to the valid unitary diagonalization already established in P04. It is not an inference from the real parts of the spectrum of an arbitrary nonnormal matrix. The explicit symmetric-part calculation confirms that no correction is needed at P09 lines 434–436.

It follows that

$$y^T \dot{y} \le -\delta_n \|y\|_2^2 + \|B\|_2 \|y\|_2^2 \le -\frac{\delta_n}{2}\|y\|_2^2.$$

8.4 Strict decrease, invariant sublevels and continuation

Choose $0 < \rho < 1/2$ so that the coordinate box $|r|, \|y\|_2 < \rho$ lies in all preceding neighborhoods and $K\rho \le \delta_n/4$. Then for $V = (r^2 + \|y\|_2^2)/2$,

$$\begin{aligned}
\dot{V} &= -\frac{Ar^4}{1+r} + rR + y^T \dot{y} \\
&\le -\frac{A}{1+\rho} r^4 + K\rho \|y\|_2^2 - \frac{\delta_n}{2}\|y\|_2^2 \\
&\le -\frac{A}{1+\rho} r^4 - \frac{\delta_n}{4}\|y\|_2^2 < 0 \quad \text{if } (r,y) \ne (0,0).
\end{aligned}$$

For negative r, $0 < 1 + r \le 1 + \rho$, so the denominator comparison has exactly the indicated direction. The term rR was bounded in absolute value; no sign assumption on r or R is used.

For example take $0 < \ell < \rho^2/4$. On the closed sublevel $V \le \ell$, $|r|, \|y\|_2 \le \sqrt{2\ell} < \rho$, and

$$|x_i| \le |r| + \|y\|_2 \le 2\sqrt{\ell} < \rho < 1.$$

This is a compact subset of $\mathcal{U}$ and of the strict-estimate neighborhood. Since $\dot{V} < 0$ on its nonzero boundary, it is positively invariant. The smooth vector field is bounded and locally Lipschitz near this

compact set, so the local solution continues for all $t \geq 0$ while remaining there. The coordinate norm is equivalent to the full state norm:

$$\frac{\|x\|_2^2}{2n} \leq V \leq \frac{\|x\|_2^2}{2}.$$

Small invariant sublevels give Lyapunov stability. Within any such compact set, the only point where $\dot{V} = 0$ is zero. The strict Lyapunov theorem, or LaSalle on that set, gives convergence to zero. This proves local asymptotic stability in the *full* open domain, without presuming global invariance of $\mathcal{U}$. P09 lines 437–449: **PASS**.

Explicit substantive conclusion: the full origin at $b = 0$, $\mu = 0$ is locally asymptotically stable in $\mathcal{U}$. VERIFIED.

8.5 Algebraic decay rules out exponential stability

On synchrony at the same tuple, a small positive solution decreases and a small negative solution increases. Uniqueness prevents either from crossing or reaching the equilibrium in finite time. Monotonicity and the absence of any other scalar equilibrium force convergence to zero. For every nonzero such solution,

$$\frac{d}{dt}(r^{-2}) = -2r^{-3}\dot{r} = \frac{2A}{1 + r(t)} \longrightarrow 2A.$$

Integrating and taking a Cesaro mean gives $r(t)^{-2}/t \to 2A$, hence $|r(t)| \sim (2At)^{-1/2}$. Sign preservation makes this valid from either side. If the full origin were locally exponentially stable, these exact full-system solutions would satisfy $\sqrt{n}|r(t)| \leq Me^{-\gamma t}\sqrt{n}|r_0|$ for some $M, \gamma > 0$. Their algebraic decay contradicts that estimate. No rate for every nonsynchronous trajectory is inferred. P09 lines 451–459: **PASS**.

DEGENERATE CRITICAL-STABILITY VERDICT: VERIFIED.

9 Layer F: codimension, terminology and scope

In the open positive parameter space with coordinates (a, k, c, g, h), set $H_1 = a-k-h$ and $H_2 = g-a-c$. Their gradients are

$$\nabla H_1 = (1, -1, 0, 0, -1), \qquad \nabla H_2 = (-1, 0, -1, 1, 0).$$

A linear dependence has second coefficient zero by the g coordinate, then first coefficient zero by the k coordinate. Equivalently the minor in columns k, g is -1. Rank two holds everywhere. The common zero set is nonempty, since any $k, c, h > 0$ give $a = k + h$, $g = a + c$. The regular level-set theorem gives a smooth codimension-two submanifold for each fixed admissible (n, q).

This is rank in physical parameter space, not algebraic multiplicity of an eigenvalue. At every point on the locus, $J_c = c(P - I)$ still has exactly one simple zero. P09 expressly excludes a double-zero spectral degeneracy; there is no Bogdanov–Takens assertion.

Verified P02 gives

$$g_f = a + c + \frac{(a - A)^2}{4A}, \qquad r_f = \frac{a - A}{2A}.$$

As $A > 0$, $g_f = g_c$ iff $a = A$, at which $r_f = 0$. The repeated root is positive only when $a > A$. Thus the meeting point is the origin and the statement is only an algebraic meeting of loci. In fact the two equations $g = g_c$ and $g = g_f$ themselves meet tangentially: the rank-two proof uses $b = 0$ and $g - g_c = 0$, not independent gradients of the two locus graphs. P09 does not confuse these two rank statements. P09 lines 462–492: **PASS**.

P09's degenerate terminology is consistently "cubic-degenerate origin", "square-root nonzero branches", and failure of ordinary transcritical quadratic nondegeneracy. Its occurrences of "pitchfork" explicitly withhold that classification. No $\mathbb{Z}_2$ symmetry is asserted. The rational scalar field is not presumed odd. No named codimension-two unfolding is claimed.

The negative degenerate stable branch is a signed equilibrium in $\mathcal{U}$; it supplies no second positive stable branch for a physical bistability theorem. The generic $b > 0$ case only analyzes the branch near zero. P09 neither proves nor claims a positive-fold saddle-node, true bistability, basin separation, basin geometry, cusp unfolding or two-parameter unfolding. P10 retains those subsequent tasks within the frozen scope; B03 still requires its precise local coexistence wording. Likewise there is no P05 selection theorem, five-node specialization, DM–MB comparison, Hopf assertion, imaginary-pair transversality, first Lyapunov coefficient, rotating wave or large-n theorem here. The real branch-crossing word "transverse" is not an imaginary-pair transversality claim. Neighborhoods may depend on n; the gap shrinks with growing n and no uniform-in-n assertion is made. P09 lines 494–539: **PASS**.

CODIMENSION / SCOPE VERDICT: VERIFIED.

10 Independent corroboration, provenance and counterexample search

10.1 Independent checks

The audit's `A09_support/independent_checks.py` was initially authored from the canonical field before reading the developer checker. It imports no developer code or output. It uses exact rational series division and rational identities separately from ordinary NumPy floating-point tests. Its final recorded run used Python 3.12.14 and NumPy 2.3.5, with fixed random seed 90261003. The source and all raw results are packaged.

Independent check class	Count
Exact rational identities and derivative checks	310
Original-field equilibrium / direct-matrix cases	2,370
Generic local slope checks, both orientations	316
Original-field transverse Newton searches	474
Full-field Lyapunov / remainder / transverse-factor samples	1,580

Dimensions are $5, 6, 7, 8, 9, 10, 12, 15, 16$, with every admissible q: 79 network pairs, including 33 noncoprime pairs. This includes $q = 1$, $q = n - 1$ and $q = n/2$ whenever allowed. Six positive fixed tuples include both generic signs, two exactly degenerate tuples, a small coupling c, and b close to zero on both sides. Both signs of μ, critical points, the generic local branch and both degenerate roots are tested.

The original field is evaluated directly. Jacobian entries are assembled by adding each summand's derivative to its actual receiving column, including collisions, and checked against complex-step differentiation. All direct matrix eigenvalues are paired to the analytic list with multiplicity retained. Maximum equilibrium residual is 1.214×10^{-20}; maximum scaled spectrum discrepancy is 4.797×10^{-15}; maximum scaled Jacobian discrepancy is 6.939×10^{-16}; maximum relative slope error is 1.251×10^{-7}. These maxima are run diagnostics, not rigorous enclosures.

The transverse searches fix (r, μ) and solve the original projected field in a real orthonormal basis of E from small nonzero initial transverse states. All converge to numerical zero; the largest final transverse norm is 1.636×10^{-19}. This is only a finite search, not a proof of IFT uniqueness. The Lyapunov tests independently evaluate R, $\dot{V}$, the symmetric-part eigenvalues, and a quadrature approximation of B.

All satisfy the tested inequalities. Tiny-b eigenvalues may approach floating-point resolution; their sign is checked by the exact scalar formula, not falsely certified by a rounded matrix eigenvalue.

10.2 Developer validation is kept separate

The developer companion was located and its original checker authenticated:

`b42af82b97c187b3a61238b7cc2aba9d35e44d210a38ec88421b10d5be2fcf93`

It was copied into the audit support directory and rerun there, leaving the original untouched. All four result files (exact, numerical, slopes, summary) reproduced byte-for-byte: 1,260 exact checks, 2,700 matrix cases and 12 slope checks. The originals, rerun files and hash comparison are under `A09_support/prior_validation/`. Their role is validation provenance only. This reproduction neither proves P09 nor resolves the older A04/A05 provenance qualifications.

10.3 Twelve targeted attempts to break the proof

No.	Target	Adversarial resolution
1	$\psi \equiv 0$	Tested parameter-dependent neighborhoods and uniqueness. A fixed product contains $y = 0$ for every nearby (r, μ); uniqueness forces zero. This is analytic, not a numerical observation.
2	Equilibrium completeness	A merely scalar restriction would be inadequate. The invertible real transverse derivative excludes every nearby nonsynchronous solution, including nonsmooth hypothetical branches.
3	Branch slope sign	Differentiate Q twice and then use $d\mu/dg = -1$; both signs of b checked. No missing orientation reversal.
4	Stability exchange	Recomputed all modes, including even singleton and offset collisions. Only the uniform eigenvalue can change sign locally.
5	Positive branch side	The exact relation $\mu = -r_*(b - Ar_*)$ fixes the side. An additional root away from zero cannot be substituted for r_*.
6	Time factor	Both potential zeros $r = b/A$ and $r = b/(2A)$ are excluded by a fixed-b neighborhood; $r = -1$ is outside the domain. $\alpha(0) = 1$.
7	Generic critical escape	For $b < 0$, negative r gives increasing $\lvert r\rvert$, rather than positive escape. The source preserves the orthant scope.
8	Negative degenerate root	$1 + r_- > 0$ locally and the uniform numerator is $-2Ar_-^2 < 0$. No positive-state coefficient signs were imported.
9	Full degenerate stability	Both the fixed-parameter reduction principle and an independent strict Lyapunov function apply; parameter neutrality cannot be used to refute the fixed-tuple assertion.
10	Mean remainder	A potential $\lvert r\rvert\lVert y\rVert$ term vanishes identically: $w^T J(r)\rvert_E = 0$ at every r. Exact formula (2) checks this.
11	Quadratic-form bound	Real parts alone would fail for a general nonnormal matrix. Here the symmetric part has exactly the Fourier cosine eigenvalues, proving the asserted bound.
12	Codimension	Independent gradients of b and $g - g_c$ have rank two; the tangential meeting of the repeated-root and intersection graphs is not mistaken for a transverse intersection.

No analytic counterexample survives these checks. No finite computation is claimed to exclude counterexamples universally; the analytic arguments in Sections 3–9 supply that justification under P09's stated local hypotheses.

11 Complete source-line review

Line numbers refer to the authenticated 539-line P09 v1.0 source, not its PDF or an earlier placeholder. The ledger below covers every line once in consecutive ranges, including metadata, separators and the closing checklist. Every substantive range is assessed. All ranges are PASS; there is no MINOR ISSUE, MAJOR ISSUE, FALSE or UNRESOLVED finding. The machine-readable companion is `A09_support/line_review.json`.

Lines	Verdict	Finding
1–15	**PASS**	Metadata correctly declares DRAFT, fixed parameter path, direct P02–P04 dependencies, inherited P01 and scope exclusions.
16–20	**PASS**	Title and status agree with the exact CURRENT manifest; no verification is self-awarded.
21–27	**PASS**	Canonical tuple and abbreviations are consistent; b is unambiguously local and $A > 0$.
28–33	**PASS**	All five original DM terms are transcribed with correct indices and $g = g_c - \mu$.
34–39	**PASS**	Joint smoothness, positive g neighborhood, open domain and parameter-dependent locality are correct; no global invariance or uniform-in-n claim.
40–49	**PASS**	P02 import is the exact invariant restriction with denominator $1 + r$; no global completeness is inferred here.
50–53	**PASS**	Subsection and anchor only; no added claim.
54–70	**PASS**	Critical spectrum lemma: algebraically simple zero, real kernel, explicit positive gap and arbitrary admissible q are correct.
71–80	**PASS**	Original differentiation, Fourier phases, characteristic-polynomial derivative and real kernel are independently correct.
81–85	**PASS**	Cosine comparison covers odd/even dimensions and all paired/singleton nonzero labels.
86–97	**PASS**	Local equilibrium completeness is stated in state and parameter neighborhoods, including $b = 0$.
98–107	**PASS**	Mean coordinates and orthogonal mean-zero projection give a real linear isomorphism; normalization agrees with scalar amplitude.
108–115	**PASS**	Projected derivative preserves E; complexification has no kernel, which does imply invertibility of the real restriction.
116–124	**PASS**	Real IFT product and uniqueness imply $\psi \equiv 0$; zero belongs to one common transverse neighborhood.
125–130	**PASS**	Both equilibrium implications are proved; multiplication by positive $1+r$ is harmless; shrinking gives a state/parameter product.
131–135	**PASS**	Subsection/proposition opening has no unsupported claim.
136–141	**PASS**	Scalar derivative values and mixed derivative are correct.
142–154	**PASS**	Left/right vectors have pairing one; full contractions, Taylor half-factor and opposite physical-g coefficient are correct.
155–159	**PASS**	Scalar expansion has cubic coefficient $-(A + b)$ and quadratic derivative $2b$.

Lines	Verdict	Finding
160–169	**PASS**	Original-field mixed derivatives and all five second directional contributions are correct; averaging introduces no extra n.
170–175	**PASS**	Generic theorem title is justified by the ensuing full-system proof.
176–189	**PASS**	Branch completeness, smoothness and expansions have fixed-b local scope; both slope orientations are correct.
190–206	**PASS**	Tangents are in the synchronous state–parameter plane; full nonlinear stability table and transcritical conclusion have the stated generic hypotheses.
207–215	**PASS**	Second IFT constructs the unique small root; nonzero iff $\mu \neq 0$; the other root stays excluded for fixed $b \neq 0$.
216–225	**PASS**	First/second derivatives and tangent vectors are correct; $d\mu/dg = -1$ is explicitly used.
226–235	**PASS**	Invariant line is a fixed-parameter center manifold; its product with parameter is the full extended center manifold because $F_\mu(0,0) = 0$.
236–244	**PASS**	Coordinate $u = Ar^2 - br$ is locally invertible; exact transformed vector field and rational time factor are correct.
245–249	**PASS**	Denominators stay nonzero, time factor is positive, and orbital equivalence is limited to the center restriction.
250–254	**PASS**	Full derivative of the scalar field is taken before imposing equilibrium; product-rule terms are correct.
255–261	**PASS**	Uniform eigenvalue and alternative expression use the equilibrium relation with the correct minus sign.
262–270	**PASS**	Both b factors retain their sign locally; the exact sign proof is primary and the expansion only corroborates it.
271–282	**PASS**	General P04 coefficients and phase signs remain valid at negative synchronous states and colliding offsets.
283–293	**PASS**	Finite-label continuity yields one common strict transverse gap, including $b = 0$; no selection theorem is used.
294–303	**PASS**	Origin sign and nonlinear linearization results prove the exchange. Exactly one unstable real direction refers to the off-critical hyperbolic equilibria just treated.
304–315	**PASS**	Physical corollary has correct branch side and stability for each sign of b; orthant statements are off-critical.
316–324	**PASS**	Positive synchronous escape, with a fixed radius for each fixed $\mu > 0$, supplies an admissible trajectory proving relative instability.
325–330	**PASS**	Nonnegative invariance restricts stability; interior positive-branch instability and the separate away-from-zero root are correctly scoped.
331–338	**PASS**	Generic critical instability is in $\mathcal{U}$; the $b < 0$ orthant conclusion is expressly withheld.
339–343	**PASS**	Critical synchronous equation is exact.
344–353	**PASS**	Magnitude bound $\lvert b\rvert r^2/3$, sign choice for r, finite exit, domain preservation until exit and orthant limitation are correct.
354–359	**PASS**	Degenerate subsection/proposition opening does not assume genericity.
360–367	**PASS**	For $\mu < 0$ and $\mu = 0$, local root counts and stability types are correct; triple zero describes the scalar field.
368–377	**PASS**	Two roots for $\mu > 0$ are in $\mathcal{U}$ locally; both are exponentially stable there; origin is unstable and only the positive root is nonnegative.

Lines	Verdict	Finding
378–382	**PASS**	Translation to g is correct; pitchfork and two-parameter unfolding are explicitly not asserted.
383–388	**PASS**	The transverse reduction applies without $b \neq 0$; both roots can be placed in a common local neighborhood with $r > -1$.
389–393	**PASS**	Critical cubic derivative is $-6A$; scalar zero multiplicity is exactly three.
394–403	**PASS**	Uniform eigenvalue is negative on both signed branches; transverse continuity and the earlier positive escape supply full off-critical conclusions.
404–415	**PASS**	The fixed-parameter center-stability theorem is stated with and satisfies all hypotheses; the neutral parameter direction is explicitly excluded.
416–418	**PASS**	Independent direct Lyapunov proof is at the fixed degenerate tuple in real mean/transverse coordinates.
419–426	**PASS**	Circulant column sums kill all linear-in-y mean terms for every r; uniform transverse Taylor remainder is $O(\|y\|_2^2)$.
427–432	**PASS**	Vanishing transverse field on synchrony gives the exact integral factorization and the displayed operator $B : E \to E$.
433–436	**PASS**	Operator norm of B shrinks to zero; the quadratic-form estimate follows from unitary Fourier diagonalization, or explicitly its real symmetric part.
437–444	**PASS**	Radius choice is feasible with a fixed K; the rR bound and denominator comparison give the strict negative derivative for both signs of r.
445–450	**PASS**	Compact sublevels lie in $\mathcal{U}$ and the estimate neighborhood, are invariant, and ensure continuation and full asymptotic stability.
451–460	**PASS**	Sign preservation and convergence justify the reciprocal-square derivative limit, algebraic asymptotic, and impossibility of full exponential stability.
461–471	**PASS**	Codimension proposition concerns the positive physical parameter space and an algebraic meeting of loci only.
472–480	**PASS**	Gradients have rank two, the regular-level theorem applies, and the locus is nonempty.
481–483	**PASS**	The center zero remains spectrally simple; no double-zero claim.
484–492	**PASS**	P02 formulas give equality of loci iff $a = A$, at root zero; positive repeated root and saddle-node scope are correctly distinguished.
493–500	**PASS**	B02 resolution is limited to P09; NB07, frozen supporting role and exclusion of unfolding are respected.
501–512	**PASS**	Analytical checks are actually in the proof; computational provenance is separate and has been reproduced, without treating validation as proof.
513–518	**PASS**	Audit-priority preamble and IFT/slope priorities accurately identify the proof obligations.
519–524	**PASS**	Coefficient and center-normal-form audit requests preserve normalization and full-system completeness.
525–528	**PASS**	Spectral and physical-side priorities explicitly include negative states, collisions and open-domain versus orthant distinctions.
529–534	**PASS**	Degenerate Lyapunov and codimension priorities are correctly scoped; no named pitchfork/cusp classification is asserted.
535–539	**PASS**	Final exclusions retain the P10 boundary and all other forbidden later-module claims; closing environment is complete.

12 Requested-check coverage

The request's checks 0–1 are addressed in Sections 1–2; checks 2–9 in Sections 3 and the verdict table; checks 10–16 in Section 4; checks 17–25 in Section 5; checks 26–27 in Section 6; checks 28–34 in Section 7; checks 35–41 in Section 8; checks 42–47 in Section 9; checks 48–49 in Section 10; and check 50 in Section 11. Checks 51–55 are implemented by the verdicts, exact-slug source, audit-only archive and closure handoff. None of these dispositions depends on compiling this document or on a developer validation result.

13 Closure handoff to the original P09 development chat

OVERALL VERDICT: VERIFIED. Required corrections: none. There are no E01, E02 or other required issue IDs to close. The original P09 development chat may perform administrative closure without changing the mathematical body of the exact audited P09 v1.0. Preserve the original audit source and its evidence, and record the current/audited hashes and verdicts in the appropriate lifecycle metadata there.

This audit chat has not edited P09, promoted it to VERIFIED in any project file, begun P10, or created a replacement full project. The audit delivery is named exactly `cyclic_network_project_A09_audit_only.zip`, with the root `cyclic_network_project/` and only A09 artifacts. The original `cyclic_network_project_P09_draft.zip` remains unchanged.

After successful correction/closure, the original P09 development chat must create the COMPLETE full-project archive:

`cyclic_network_project_P09_audited.zip`.

That archive must not be created in this audit chat.

P10 gate: cleared after administrative A09/P09 audit closure. All mathematical prerequisites P02, P03, P04 and the audited P09 result are available; P10 is the next frozen attempt. This is permission to begin its scoped proof work after closure, not a P10 theorem verdict. P10 must state its own positive-equilibrium, full-spectrum and local stable-coexistence hypotheses, retain the B03/local-versus-global boundary, and undergo its own audit. No global basin result, unfolding or later proof is authorized or supplied by this audit.